27213

The Building Process

Advanced Carpentry

Overview

Construction is about much more than just building something. It includes all the steps from idea through finished product. Every construction project requires planning, budgeting, code compliance, building, and oversight to ensure its successful completion. This module examines the codes and standards that apply to construction projects, the various types of construction projects, the people involved in construction projects, and the stages of the building process for construction projects.

Learning Objective 1

Successful completion of this module prepares trainees to:

Understand the primary codes and standards used in the construction industry.

a. Identify the primary codes used in the construction industry.

b. Identify the standards that apply to the construction industry.

Learning Objective 2

Successful completion of this module prepares trainees to:

Explain the primary functions within the construction industry.

a. Describe the construction sectors.

b. Describe the roles of individuals in the construction industry.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Describe the construction process.

a. Explain the stages of the construction process.

Performance Tasks

This is a knowledge-based module. There are no Performance Tasks.

Recommended Teaching Time: 22.5 hours

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27402

Site Preparation

Advanced Carpentry

Overview

While Module 27114, *Principles of Site and Building Layout*, described the basic concepts of site layout, this module covers more advanced site preparation topics, such as setting up and checking leveling instruments, and performing specific steps associated with differential leveling. Additionally, detailed steps are provided for measuring horizontal, vertical, and traverse angles. This information, along with hands-on experience you will gain in lab and jobsite settings, will help you accurately perform these advanced site layout activities.

Learning Objective 1

Successful completion of this module prepares trainees to:

Describe how to set up and check instruments associated with differential leveling.

a. List safety hazards associated with site layout.

b. Summarize how to set up and check leveling instruments.

Learning Objective 2

Successful completion of this module prepares trainees to:

Explain how to determine site and building elevations.

a. Describe how to use a leveling instrument to calculate a site's elevations.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Describe applications involving differential leveling.

a. Explain how to transfer an elevation up a structure.

b. Summarize applications for profile, cross-section, and grid leveling.

Learning Objective 4

Successful completion of this module prepares trainees to:

Lay out building lines using radial layout techniques.

a. Explain how to measure distances by taping.

b. Describe how to measure horizontal and vertical angles.

c. Explain how to lay out building lines.

Performance Tasks

1. Set up, adjust, and field check leveling instruments.
2. Read transit level/theodolite scales and verniers.
3. Use a leveling instrument to calculate a site's elevations.
4. Record differential leveling data in field notes in accordance with accepted procedures.
5. Use differential leveling procedures to transfer elevations up a structure.
6. Use a transit level to lay out building lines using radial layout techniques.
7. Perform calculations pertaining to angular measurements.

Recommended Teaching Time: 22.5 hours

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27404

Wall Systems and Installations

Advanced Carpentry

Overview

Advanced wall systems include structural insulated panels, masonry veneer systems, fire-resistance-rated (FRR) construction, and shear walls. Although many basic construction concepts apply to advanced wall systems, additional knowledge is required to understand insulative characteristics, details about masonry veneers, wall fire protection, and the effects of lateral forces on buildings.

Learning Objective 1

Successful completion of this module prepares trainees to:

Identify safety hazards to consider when installing an advanced wall system.

a. Describe hazards that may be present when installing wall systems.

Learning Objective 2

Successful completion of this module prepares trainees to:

Describe the different types of advanced wall systems.

a. Describe structural insulated panels (SIPs) and explain the carpenter’s role in their installation.

b. Identify the carpenter’s role in the installation of masonry veneer walls.

c. Define fire-resistance-rated construction and explain methods used to develop wall systems that protect occupants from smoke and fire.

d. Describe how shear walls reinforce buildings against lateral forces.

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Performance Tasks

1. Construct a shear wall for light-frame construction.

2. Construct firewalls in accordance with specifications.

Recommended Teaching Time: 22.5 hours

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27202

Roofing Applications

Advanced Carpentry

Overview

As you travel, take note of various types of residential and commercial structures. Note the kinds of roof construction, and how many different types of roofing materials are used. Part of your work as a carpenter will involve preparing roof decks to receive the finish roofing material, and you may even install roofing material. The roof is the most vulnerable part of a building. If not properly installed, it will leak. In some situations, it could even collapse. Safety is always a major consideration when working on a roof.

Learning Objective 1

Successful completion of this module prepares trainees to:

Explain the safety requirements for roofing projects.

a. Identify potential hazards when working on roofs.

b. Identify proper personal protective equipment (PPE) and hazard control devices used when working on roofs.

Learning Objective 2

Successful completion of this module prepares trainees to:

Identify the tools and fasteners used in roofing.

a. Identify the hand tools used when working on roofing projects.

b. Identify the power tools used when working on roofing projects.

c. Identify fasteners used on roofing projects.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Identify the different roofing systems and their associated materials.

a. Identify composition shingles and their applications.

b. Identify tile/slate roofing materials and their applications.

c. Identify metal roofing and its applications.

d. Identify built-up roofing and its applications.

e. Identify single-ply roofing and its applications.

f. Explain the purpose of underlayment and waterproof membrane.

g. Discuss the purpose of drip edge, flashing, and roof ventilation.

Learning Objective 4

Successful completion of this module prepares trainees to:

Describe the installation techniques for common roofing systems.

a. Describe how to properly prepare a roof deck.

b. Explain how to install composition shingles.

c. Explain how to install metal roofing.

d. Discuss roof projections, flashing, and ventilation.

Learning Objective 5

Successful completion of this module prepares trainees to:

Describe the estimating procedure for roofing projects.

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Performance Tasks

1. Demonstrate how to install composition shingles on a specified roof and valley.
2. Demonstrate the method to properly cut and install the ridge cap using composition singles.
3. Lay out, cut, and install a cricket or saddle.
4. Demonstrate the techniques for installing other selected types of roofing materials.

Recommended Teaching Time: 22.5 hours

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27205

Steel Framing Systems

Advanced Carpentry

Overview

This module describes the uses and installation of cold-formed steel framing systems. In commercial and multifamily residential construction, it is common to use steel framing materials in place of wood studs to frame walls and partitions. It is also becoming more common in single-family residential construction as the price of lumber rises. Steel framing requires a carpenter to master tools and joining techniques different from those required in wood framing construction. Learn how to effectively complete various steel framing tasks, such as the installation of load-bearing steel assemblies to nonstructural walls, by familiarizing yourself with common steel framing materials and building methods.

Learning Objective 1

Successful completion of this module prepares trainees to:

Identify the tools and components of cold-formed steel framing systems and their safe use.

a. Identify safety guidelines that should be followed when working with steel framing

b. systems.

c. Identify steel framing materials.

d. List common steel framing tools and fasteners.

e. Explain how to perform a material takeoff for a steel frame project.

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Learning Objective 2

Successful completion of this module prepares trainees to:

Identify the steps required to lay out and install a steel stud wall.

a. Describe basic steel construction methods.

b. Explain how to frame nonstructural steel walls.

c. Describe how to frame structural steel walls.

Learning Objective 3

Successful completion of this module prepares trainees to:

Identify other steel framing applications.

a. Explain how to use steel framing members in floor and roof construction.

b. Describe how to use steel framing members in ceiling construction.

Performance Tasks

1. Estimate the amount of materials required to complete an instructor-specified steel framing project.
2. Lay out a steel stud wall with openings to include bracing and blocking.
3. Demonstrate the ability to build headers (back-to-back, box, and L-header).

Recommended Teaching Time: 22.5 hours

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27203

Thermal and Moisture Protection

Advanced Carpentry

Overview

A properly insulated building creates an environment comfortable to live or work in and economical to heat or cool. Without proper insulation, warm air will escape the building in cold weather, causing the heating system to operate constantly. This results in an increased use of energy. In hot weather, the air conditioning system will have to work harder, producing the same result. Proper insulation in walls, floors, and roof decks will minimize this problem. Vapor retarders help prevent moisture from penetrating the building. Moisture can cause serious problems, including wood decay and mold growth. A skilled craftworker knows how to select and install insulating materials and vapor retarders.

Learning Objective 1

Successful completion of this module prepares trainees to:

Describe insulation and explain why it is used in buildings.

a. Explain what insulation does.

b. Describe insulation requirements.

Learning Objective 2

Successful completion of this module prepares trainees to:

Describe insulation materials and types.

a. Define flexible insulation and explain how it is used.

b. Define loose-fill insulation and explain how it is used.

c. Define rigid or semi-rigid insulation and explain how it is used.

d. Define reflective, spray, and foam insulation and lightweight aggregates.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Explain how to install different types of insulation.

a. Describe how to install flexible insulation.

b. Describe how to install loose-fill insulation.

c. Describe how to install rigid or semi-rigid insulation.

d. Describe how to install foam insulation.

Learning Objective 4

Successful completion of this module prepares trainees to:

Explain how moisture control is accomplished in the construction of a building.

a. Explain how to design a structure to control moisture.

b. Describe how ventilation and vapor retarders are used for moisture control and explain how they are installed.

Learning Objective 5

Successful completion of this module prepares trainees to:

Explain how water and air infiltration control are accomplished in the construction of a building.

a. Describe how water-resistive barriers are used to resist liquid water.

b. Describe how air barriers are used and installed.

c. Explain how to protect openings, penetrations, and joints.

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Performance Tasks

1. Install blanket insulation in a wall.
2. Install a vapor retarder on a wall.
3. Install water-resistive barriers and materials.

Recommended Teaching Time: 22.5 hours

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45104

Drywall Installation

Drywall

Overview

Gypsum board is the most common drywall finish used inside residential and commercial construction. A variety of gypsum materials are used for these applications, and a number of construction methods are used to build standard ceilings and walls, as well as assemblies that meet building codes for fire resistance and sound control. Gypsum board drywall installation requires the knowledge and use of specialized tools, fasteners, and construction methods dictated by building codes, and therefore must be carefully considered.

While Module 27114, *Principles of Site and Building Layout*, described the basic concepts of site layout, this module covers more advanced site preparation topics, such as setting up and checking leveling instruments, and performing specific steps associated with differential leveling. Additionally, detailed steps are provided for measuring horizontal, vertical, and traverse angles. This information, along with hands-on experience you will gain in lab and jobsite settings, will help you accurately perform these advanced site layout activities.

Learning Objective 1

Successful completion of this module prepares trainees to:

Identify and describe the materials and tools required to install drywall.

a. List characteristics of gypsum board and explain why it is used.

b. Describe types of drywall fasteners and explain how they are used.

c. Identify and describe types of adhesives and explain how they are used for drywall work.

d. Identify and describe tools used to install drywall.

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Learning Objective 2

Successful completion of this module prepares trainees to:

Explain how to determine site and building elevations.

a. Describe the procedure for single- and multi-ply installations of gypsum board on ceilings and walls.

b. Identify and describe the types of drywall trims.

c. Explain how moisture is controlled to prevent its effects on walls and ceilings.

d. Explain guidelines for fire-resistance-rated and sound-rated walls.

Performance Tasks

1. Install gypsum drywall panels on wood joist ceilings and wood stud walls using the following:
    - Nails
    - Screws
2. Install gypsum drywall panels on steel grid ceilings and steel stud walls using screws.

Recommended Teaching Time: 22.5 hours

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27215

Interior Finish, Trim, and Cabinets

Advanced Carpentry

Overview

Skilled finish carpenters are always in demand for installing wall and ceiling moldings, as well as finish trim around doors and windows. These skilled craft professionals may also be called upon to install cabinets and other casework components. Quality trim and cabinetry work require an exceptional ability to measure accurately, calculate angles, and make precise, clean cuts. These skills are vital because even small errors will be visible in the finished product. Therefore, an accomplished finish carpenter must develop a reputation for reliable results and remain vigilant throughout the project to quickly address minor mistakes and make adjustments to this highly detailed work.

Learning Objective 1

Successful completion of this module prepares trainees to:

Describe the job preparation and safety planning required to complete a finish carpentry project.

a. Identify tool and equipment safety guidelines for finish carpentry tools and materials.

b. Explain how to perform a material takeoff for window, door, and ceiling trim.

Learning Objective 2

Successful completion of this module prepares trainees to:

Identify the different types of standard moldings and materials.

a. Identify the different types of base, wall, and ceiling moldings.

b. Identify the different types of window and door trim.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Explain how to install different types of molding and trim.

a. Describe common techniques for cutting and fastening trim.

b. Explain the process for installing door and window trim.

c. Describe the steps for installing base and ceiling molding.

Learning Objective 4

Successful completion of this module prepares trainees to:

Identify common types of cabinets and countertops.

a. Identify common types of wall, base, and island cabinets.

b. Describe common types of countertops.

Learning Objective 5

Successful completion of this module prepares trainees to:

Identify common cabinet components and hardware.

a. Describe various cabinet components, such as doors, drawers, and shelves.

b. Identify common fasteners and hardware used in cabinet assembly.

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Learning Objective 6

Successful completion of this module prepares trainees to:

Explain how to lay out and install a basic set of cabinets.

a. Describe common surface preparation techniques required before cabinet installation.

b. Explain how to install wall cabinets.

c. Describe the process for installing base cabinets and countertops.

Performance Tasks

1. Make square and miter cuts to selected moldings using a hand miter box and a compound miter saw.
2. Make a coped joint using a jigsaw or coping saw.
3. Install interior door, window, base, and ceiling trim using a finish nailer.
4. Identify and install selected hardware on cabinet doors.
5. Identify and lay out various types of base and wall units following a specified layout scheme.
6. Install a selected type of cabinet and countertop.

Recommended Teaching Time: 22.5 hours

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27208

Doors and Door Hardware

Advanced Carpentry

Overview

The installation of interior and exterior doors and their hardware is a specialized skill that requires careful measuring and skillful use of tools. Some doors are prehung in a frame and require accurate installation to keep the frame level and plumb. In other cases, a carpenter must learn how to install hinges to hang the door in a finished opening. Both skills will be useful to learn and master as you expand your carpentry skill set.

Learning Objective 1

Successful completion of this module prepares trainees to:

Describe the safety hazards related to working with doors.

a. Explain the common safety guidelines for handling and installing doors.

Learning Objective 2

Successful completion of this module prepares trainees to:

Identify common door frames and their installation methods.

a. Discuss common wood door frames and their installation methods.

b. Discuss common metal door frames and their installation methods.

c. Explain the importance of door swing.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Identify the different types of residential doors, their hardware, and their common installation techniques.

a. Describe the different types of residential doors.

b. Identify common residential door hardware and its applications.

c. Explain the basic steps for installing residential doors and door hardware.

Learning Objective 4

Successful completion of this module prepares trainees to:

Identify commercial doors, their hardware, and their common installation techniques.

a. Identify common commercial door types.

b. Identify common commercial door hardware and its applications.

c. Explain the basic steps for installing commercial doors and door hardware.

Performance Tasks

1. Install a metal door frame in an instructor-selected rough opening.
2. Lay out and install an instructor-selected set of hinges.
3. Lay out and install an instructor-selected lockset.

Recommended Teaching Time: 22.5 hours

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27209

Suspended and Acoustical Ceilings

Advanced Carpentry

Overview

Suspended ceilings are found in most commercial buildings. Unlike fixed ceilings, they provide easy access to wiring, cabling, and air conditioning equipment located in the area between the ceiling and the overhead deck. The ceiling tiles suppress sound transmission. Suspended ceilings are built by first installing a grid suspended from the overhead deck, and then installing ceiling tiles in the grid. Proper installation of these ceilings is a skill that takes training and practice.

Learning Objective 1

Successful completion of this module prepares trainees to:

Identify the components necessary to properly install a suspended ceiling system.

a. Identify the suspension systems and hardware necessary to properly install a suspended ceiling system.

b. Identify the system components necessary to properly frame a suspended ceiling system.

c. Identify the safe material handling and storage procedures required when installing a suspended ceiling system.

Learning Objective 2

Successful completion of this module prepares trainees to:

Interpret a reflected ceiling plan.

a. Interpret the layout information.

b. Interpret the mechanical, electrical, and plumbing (MEP) locations.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Describe the key considerations, methods, and best practices relating to ceiling installation.

a. Explain seismic considerations for ceilings.

b. Identify the layout and takeoff procedures to install a suspended ceiling system.

c. Identify the tools and equipment to lay out and install a suspended ceiling system.

d. Identify the installation methods and procedures for a suspended ceiling system.

Performance Tasks

1. Estimate the quantities of materials needed to install a lay-in suspended ceiling system in a typical room from an instructor-supplied drawing.

2. Establish a level line at ceiling level, such as is required when installing the wall angle for a suspended ceiling.

3. Lay out and install a lay-in suspended ceiling system according to an instructor-supplied drawing.

Recommended Teaching Time: 22.5 hours

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27405

Finished Stairs

Advanced Carpentry

Overview

Stairways represent an important part of the interior design of a structure. A great deal of finish work needs to be done to transform a basic set of stairs into an attractive stairway.

Observe the following guidelines when installing any stairway components or a balustrade system:

Learning Objective 1

Successful completion of this module prepares trainees to:

Examine the procedure for installing various stairway systems.

a. Explain the procedure for cutting and installing treads and risers for closed stairways.

b. Identify the procedure for installing open, combination, L-shaped, and U-shaped stairways.

c. Describe the procedure for installing handrails, guards, and balustrades.

Learning Objective 2

Successful completion of this module prepares trainees to:

Describe the procedure for installing custom stairway systems.

a. Outline the procedure for installing shop-built stairways.

b. Describe the procedure for installing winder treads.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Summarize the process of building exterior stairways.

a. Describe the procedure for building exterior wood stairways.

b. Explain the procedure for building cast-in-place forms for concrete stairways.

Performance Tasks

1. Install treads and risers on open, closed, and/or combination open/closed main stairways.
2. Miter a finished stringer and risers.
3. Install a return nosing.
4. Install a post-to-post balustrade system.
5. Build a cast-in-place form for a concrete stairway.

Recommended Teaching Time: 22.5 hours

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27204

Exterior Finish and Trim

Advanced Carpentry

Overview

There are a wide variety of finishing materials used to cover the exteriors of homes and commercial buildings, including wood, brick, vinyl, metal, and fiber-cement. Each type of material has its own preparation requirements and installation practices. These materials produce an aesthetic quality, but they also protect the building from the elements and seal the building at roof overhangs with soffit and fascia. Always follow the engineer's and manufacturer's instructions to maintain warranties and improve the longevity of your project.

Learning Objective 1

Successful completion of this module prepares trainees to:

Explain the common hazards you must identify and mitigate in your site safety plan when installing exterior trim and finish materials.

a. Describe common safety hazards of working from ladders, elevated platforms, and aerial lifts.

b. Discuss common safety hazards of exterior finishing tools and materials.

Learning Objective 2

Successful completion of this module prepares trainees to:

Explain the steps to prepare the exterior for siding installation.

a. Discuss which construction documents will provide useful information for the exterior.

b. Explain the basic concepts of performing a material takeoff for siding materials.

c. Discuss common types of insulation, flashing, furring, and building wrap that might be applied to exterior walls.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Discuss common types of exterior siding materials and their installation processes.

a. Describe the basic principles of exterior siding layout.

b. Discuss common siding styles and their installation methods.

c. Identify exterior trim components including soffits, fascia, and cornices.

Performance Tasks

1. Perform a wall inspection, and apply flashing, furring, insulation, and building wrap to prepare the wall surface for exterior siding.

2. Lay out and install three of the most common siding types in your area.

Recommended Teaching Time: 22.5 hours

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27214

Mass Timber Construction

Advanced Carpentry

Overview

Mass timber construction has become increasingly popular in the US. More companies are manufacturing mass timber products and constructing mass timber buildings than ever before. To go along with this increase, there have been many code changes specific to mass timber. Mass timber products can be used in tall buildings, schools, hotels, multifamily residential buildings, public/institution buildings, and healthcare facilities. Because most mass timber products are prefabricated, there are specific guidelines for transportation, handling, and installation. This module will give you an overview of mass timber codes, types of buildings, benefits of mass timber, and framing systems used with mass timber construction.

Learning Objective 1

Successful completion of this module prepares trainees to:

Describe the use of mass timber in the construction industry.

a. Identify the building codes related to the use of mass timber.

b. Examine the benefits of mass timber in different types of construction.

c. Categorize the types of mass timber.

Learning Objective 2

Successful completion of this module prepares trainees to:

Summarize building techniques and tools used with mass timber.

a. Explain specific handling, tool, and hardware requirements for mass timber.

b. Recognize mass timber framing systems.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Describe common construction phases in the mass timber building process.

a. Provide a brief explanation of the importance of early design phases in mass timber construction.

b. Describe common safety considerations during the construction phase.

c. Explain mass timber finishing techniques used during the post-construction phase.

Performance Tasks

This is a knowledge-based module. There are no Performance Tasks.

Recommended Teaching Time: 22.5 hours

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46100

Introduction to Leadership

Introduction to Leadership

Overview

As a frontline leader, you must be able to communicate effectively, provide direction to your crew, plan and schedule work, manage costs, and ensure the jobsite is safe for everyone. Whether you are already a frontline leader or want to become one, this module will help you learn more about the requirements and skills needed to succeed.

Learning Objective 1

Successful completion of this module prepares trainees to:

Explain business and organizational structures in the construction industry.

a. Understand organizational charts.

b. Describe organizational roles and relationships, including job descriptions and responsibility, authority, and accountability.

Learning Objective 2

Successful completion of this module prepares trainees to:

Describe frontline leadership and communication skills.

a. Describe the qualities and role of a leader, including leadership styles and ethical leadership.

b. Explain the communication process and distinguish between verbal, nonverbal, written, and visual communication types.

c. Compare problem-solving and decision-making.

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Learning Objective 3

Successful completion of this module prepares trainees to:

Identify key skills for team leadership and motivation.

a. Choose strategies to motivate individuals and a team.

b. Describe key approaches to building and leading teams.

c. Recognize how to approach leadership challenges, such as transitioning from peer to leader, addressing performance issues, and resolving conflict.

Learning Objective 4

Successful completion of this module prepares trainees to:

Identify a frontline leader's safety leadership responsibilities.

a. Assess the impact of incidents.

b. Explain the purpose of OSHA and describe the role of OSHA in administering worker safety.

c. Describe the importance of safety and health programs.

d. the frontline leader's role in incident preparedness and response.

e. Recognize the importance of supporting your crew's total health.

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Learning Objective 5

Successful completion of this module prepares trainees to:

Demonstrate an understanding of construction project phases, project planning and scheduling, field reporting, and cost and resource management.

a. Identify construction project phases.

b. Summarize crew planning and scheduling responsibilities, including creating lookahead schedules.

c. Describe the frontline leader's role in field reporting.

d. Explain approaches to managing costs and controlling project resources, such as materials, tools, and equipment.

Performance Tasks

1. Organize and deliver a safety-related toolbox talk, applying communication skills.

2. Complete a job safety analysis.

3. Develop and present a lookahead schedule.

Recommended Teaching Time: 22.5 hours

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